GEOPHYSICAL : FIELDS

EQUATIONS

PHYSICAL PROPERTIES

MIXING LAWS

2017 1

WHAT IS A FIELD?

ANYTHING WE CAN DESCRIBE ~~AS A FIELD OF POINTS~~ IN SPACE AS HAVING A POSITION, AND A MAGNITUDE AND/OR DIRECTION.

EXAMPLES:

TEMPERATURE (SCALAR FIELD) - MAGNITUDE ONLY

ELEVATION " "

GRAVITATIONAL FORCE (VECTOR FIELD) - MAGNITUDE + DIRECTION (MONOPOLAR)

WIND VELOCITY " "

MAGNETIC FIELD " " (DIPOLAR)

STRESS (TENSOR FIELD) - MAGNITUDE + DIRECTION (HIGHER ORDER THAN VECTOR FIELD)

STRAIN " "

PRESSURE (SAME UNITS AS STRESS, BUT ~~XXX~~ SCALAR MAGNITUDE ONLY)

COMPOSITION (SCALAR FIELD). MULTIPLE MAGNITUDES.

WE CAN USE A FUNCTION TO DESCRIBE A GEOPHYSICAL FIELD

FOR EVERY POSITION x, $f(x)$ produces a single value y

function of x

$$x \mapsto f(x) = y$$

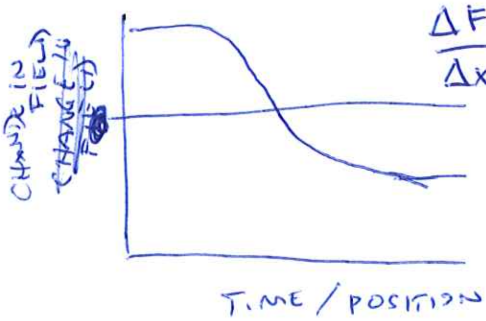
GEOPHYSICAL EQUATIONS:

WE WANT TO KNOW WHERE, WHY AND HOW FIELDS CHANGE.

IF THEY DON'T CHANGE (BORING) THE WORLD IS CONSTANT.

FIELDS CAN CHANGE IN <u>SPACE</u> OR <u>TIME</u> IN RESPONSE TO THE DISTRIBUTION OF OR MODIFICATION BY PHYSICAL PROPERTIES.

HOW TO ~~MEASURE~~ DESCRIBE A CHANGING FIELD



$\frac{\Delta F}{\Delta x}$ or $\frac{\Delta F}{\Delta t}$

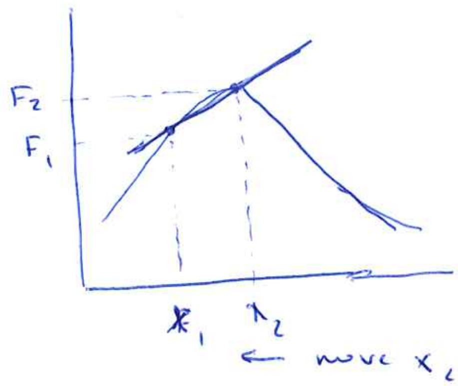
Change in $\Delta t = t_2 - t_1$

some time later — some time

$F(x)$ or $F(t)$ $\Rightarrow$ $f(x)$ or $f(t)$

FUNCTION — ALSO A FUNCTION

$$\frac{\Delta F}{\Delta x} = \frac{F(x_2) - F(x_1)}{x_2 - x_1} \rightarrow \text{slope}$$



← MOVE x_2 TOWARDS x_1 TO IMPROVE ESTIMATE OF $\frac{\Delta F}{\Delta x}$ at x_1 $\Rightarrow$ derivative

TO WRITE THIS IN 3-D

$F(x, y, z)$

x - direction   z - direction

$$\left(\frac{\Delta F}{\Delta x}, \frac{\Delta F}{\Delta y}, \frac{\Delta F}{\Delta z} \right) \Rightarrow 3 \text{ components}$$

y - direction

partial slope

TO WRITE AS A DERIVATIVE

x - direction   y - direction   z - direction

$$\left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) \text{ or as } \nabla F = \frac{\partial F}{\partial x}\hat{x} + \frac{\partial F}{\partial y}\hat{y} + \frac{\partial F}{\partial z}\hat{z}$$

gradient operator

a '+' but cannot add components.

~~[illegible]~~

GEOPHYSICAL EQUATIONS

$\nabla^2 \phi = 0$, no sources

$\nabla^2 \phi = s(x,y,z)$ with sources.

POTENTIAL FIELDS EQUATIONS
2^{nd} ORDER CHANGES IN SPACE
NO CHANGE IN TIME
(steady-state)

$\nabla^2 = \nabla \cdot \nabla$ - change in slope, curvature

field, slope, curvature example:

HIT A BALL ITS POSITION IN TIME IS THE FIELD, ITS VELOCITY (CHANGE IN POSITION WITH TIME) IS THE SLOPE AND ACCELERATION (CHANGE IN VELOCITY WITH TIME) IS CURVATURE

EXAMPLE 2:

TEMPERATURE AT A GIVEN TIME

CHANGE IN TEMPERATURE WITH TIME (GETTING HOTTER OR COLDER?)

HOW FAST IS IT GETTING HOTTER OR COLDER? THINK RECENT CLIMATE CHANGE.

WE WILL SOLVE MANY OF OUR PROBLEMS IN 1-D

GEOPHYSICAL EQUATIONS.

WE WILL FOCUS ON THREE:

POTENTIAL FIELDS

GENERAL FORM

$\nabla^2 \phi = 0$ LAPLACE'S EQN

$\nabla^2 \phi = s$ POISSON'S EQN

GEOPHYSICAL SUBJECTS

GRAVITY

$\nabla^2 U = 4\pi G \rho_m$

gravitational constant: G

gravitational potential: U

mass density: ρ_m

MAGNETICS

$\nabla^2 A = -\mu \nabla \cdot \vec{M}$

magnetic perm.: μ

magnetization: $\vec{M}$

THERMAL / GEOTHERMICS

$\nabla^2 T = -\frac{A}{k}$

heat production: A

thermal conductivity: k

temperature: T

DIFFUSION

$D \nabla^2 \phi = \frac{\partial \phi}{\partial t}$

diffusion constant: D

time variable: t

Thermal

$\kappa \nabla^2 T = \frac{\partial T}{\partial t}$

thermal diffusivity: κ

Electrical

$\nabla^2 E = \mu \sigma \frac{\partial E}{\partial t}$

electric field: E

conductivity: σ

Magnetic

$\nabla^2 H = \mu \sigma \frac{\partial H}{\partial t}$

magnetic field: H

WAVES

~~velocity of medium²~~

$\frac{\partial^2 \phi}{\partial t^2} = v^2 \nabla^2 \phi$

velocity of medium: v

P-waves

$\frac{\partial^2 \phi}{\partial t^2} = V_p^2 \nabla^2 \phi$

dilation field: ϕ

S-waves

$\frac{\partial^2 \psi}{\partial t^2} = V_s^2 \nabla^2 \psi$

GEOPHYSICAL PROPERTIES

SOME OF THE PROPERTIES WE WILL SEE IN THE COURSE

STATE VARIABLES

PRESSURE (P) units MPa, GPa, kbar

TEMPERATURE (T) K or °C

Remember a difference in T (ΔT) is the same for °C and K so no need to convert units of difference.

Composition

Volume (V) m^3 or km^3

Density

Density - mass per unit volume

$$\rho = \frac{m}{V} \qquad \text{units } kg/m^3$$

Expansivity

Expansivity - expansion due to ~~volume~~ heating

$$\alpha_V = \frac{1}{V}\left(\frac{\partial V}{\partial t}\right)_P \approx \frac{1}{V}\left(\frac{\Delta V}{\Delta T}\right)_P \qquad \text{units } K^{-1}$$

volumetric

at constant pressure

COMPRESSIBILITY volume change resulting from increase in pressure

$$\beta = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = K_T^{-1}$$

isothermal bulk modulus.

constant temperature

K_T - units MPa or GPa. LARGER value = stronger.

$K_S = K_T[1 + \gamma_{th}\alpha_v T]$ - isentropic bulk modulus

"constant entropy"

Grüneisen parameter.

SHEAR MODULUS.

μ - resistance to shear

Seismic velocity

P-waves $V_p = \sqrt{\dfrac{K_S + \frac{4}{3}\mu}{\rho}}$

S-waves $V_s = \sqrt{\dfrac{\mu}{\rho}}$

ROCKS ARE MIXTURES OF MINERALS IN various PROPORTIONS.

TO ESTIMATE BULK PROPERTIES FROM KNOWN MINERALOGY, WE MUST ~~...~~ APPLY A MIXING RELATION.

ARITHMETIC MEAN

~~weighted by~~

WEIGHTED BY MASS FRACTION OR VOLUME FRACTION OR MOLE FRACTION

– DEPENDS ON PROPERTY.

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} = w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$

mean value

denotes sum from constituent 'i' to 'n'

PROPERTIES: Density, specific heat capacity.

GEOMETRIC MEAN.

$$\bar{x} = \left(\prod_{i=1}^{n} x_i^{w_i} \right)^{\frac{1}{\sum_{i=1}^{n} w_i}} = \left[x_1^{w_1} x_2^{w_2} x_3^{w_3} \ldots x_n^{w_n} \right]^{\frac{1}{w_1 + w_2 + w_3 + \ldots w_n}}$$

product

PROPERTIES: conductivity of random mixture (esp. thermal conductivity.)

HARMONIC MEAN

$$\bar{X} = \frac{\sum_{i=1}^{n} w_i}{\sum_{i=1}^{n} \frac{w_i}{X_i}}$$

PROPERTIES (LAYERED Thermal conductivity)

VOIGT - REUSS - HILL AVERAGE

$$\bar{X} = \frac{1}{2}\left(X_V + X_R\right)$$

average of Voigt and Reuss averages.

$$X_V = \frac{\sum_{i=1}^{n} w_i v_i}{\sum_{i=1}^{n} w_i}$$

Voigt average. is arithmetic mean

$$X_R = \frac{\sum_{i=1}^{n} w_i}{\sum_{i=1}^{n} \frac{w_i}{x_i}}$$

Reuss average is harmonic average.

properties : Seismic velocity.